

Enterprise Server Deployment and Operating System Installation

Installation Guide

Course: ITEP 414 – System Administration and Maintenance

Week 3 Portfolio Project

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Program: BSIT 4C

1. Project Overview

This project covers the deployment of an enterprise Linux server using Ubuntu Server, done in the role of a Junior System Administrator setting up infrastructure for a fictional startup, TechLine Solutions. The server was installed, configured, secured with SSH access, and verified for functionality. In addition, a Windows Server Evaluation installation was completed for comparison purposes, along with a full breakdown of BIOS versus UEFI boot technology and an operating system comparison between Windows Server, Ubuntu Server, and Rocky Linux.

2. Objectives

- Explain the purpose of an operating system in enterprise environments
- Differentiate BIOS and UEFI firmware
- Explain the stages of the computer boot process
- Compare Ubuntu Server, Windows Server, and Rocky Linux
- Install Ubuntu Server in a virtual machine
- Configure server settings during installation
- Enable secure remote administration using SSH
- Verify server functionality
- Document installation procedures

3. Software Requirements

- Oracle VirtualBox
- Ubuntu Server 26.04 LTS ISO
- Windows Server 2025 Evaluation ISO

4. Virtual Machine Specifications

Component	Specification
Name	Ubuntu-Server-Week03
RAM	4 GB
CPU	2 Virtual Processors
Storage	40 GB (VDI)
Network	NAT
Optical Drive	Ubuntu Server 26.04 LTS ISO

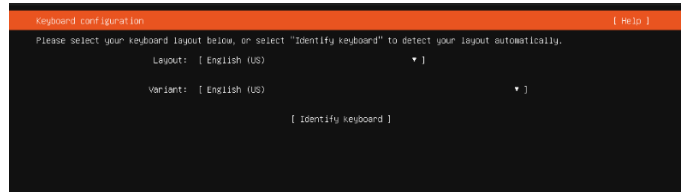
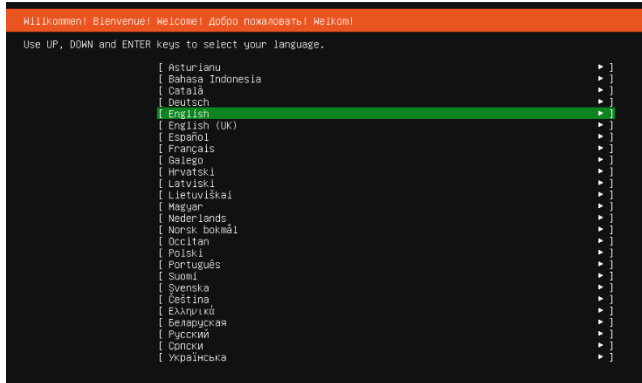
5. Step-by-Step Installation Procedure (Ubuntu Server)

Step 1: Language Selection

English was selected as the installation language.

Step 2: Keyboard Layout

Keyboard layout was set to English (US).

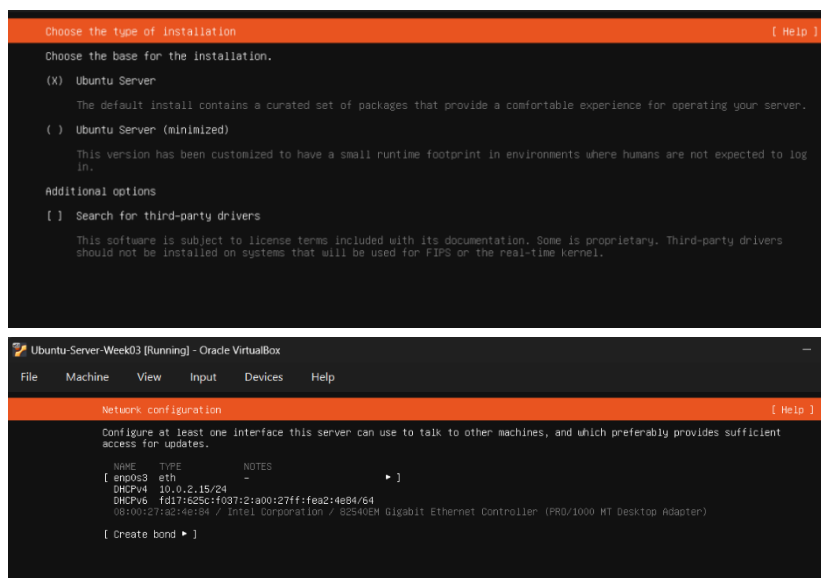


Step 3: Installation Type

Ubuntu Server (standard, not minimized) was selected.

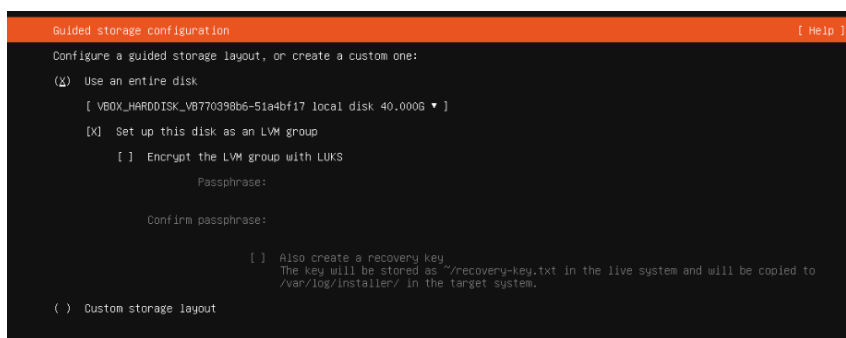
Step 4: Network Configuration

Network was configured automatically through DHCP, and the server was assigned IP address 10.0.2.15.



Step 5: Disk Partitioning

Guided storage configuration was used, selecting "Use an entire disk" with LVM enabled.



Step 6: Storage Summary

The system generated a partition summary before writing to disk: a 2 GB /boot partition (ext4) and the remaining space allocated as an LVM volume group for the root partition.

```
Storage configuration

FILE SYSTEM SUMMARY

MOUNT POINT      SIZE      TYPE      DEVICE TYPE
[ /               18.996G   new ext4   new LVM logical volume ▶ ]
[ /boot          2.000G   new ext4   new partition of local disk ▶ ]

AVAILABLE DEVICES

DEVICE              TYPE              SIZE
[ ubuntu-vg (new)    LVM volume group  37.996G ▶ ]
free space          19.000G ▶

[ Create software RAID (md) ▶ ]
[ Create volume group (LVM) ▶ ]

USED DEVICES

DEVICE              TYPE              SIZE
[ ubuntu-vg (new)    LVM volume group  37.996G ▶ ]
ubuntu-lv           new, to be formatted as ext4, mounted at / 18.996G ▶

[ VBOX_HARDDISK_VB776998b6-51a4bf17      local disk  40.000G ▶ ]
partition 1 new, BIOS grub spacer          1.000M ▶
partition 2 new, to be formatted as ext4, mounted at /boot 2.000G ▶
partition 3 new, PV of LVM volume group ubuntu-vg 37.997G ▶
```

Step 7: SSH Configuration

OpenSSH Server was installed during setup with password authentication enabled, allowing secure remote access.

```
SSH configuration

You can choose to install the OpenSSH server package to enable secure remote access to your server.

[X] Install OpenSSH server

[X] Allow password authentication over SSH

[ Import SSH key ▶ ]

AUTHORIZED KEYS

No authorized key
```

Step 8: Installation Progress

The installer proceeded to partition the disk, extract the system files, and install packages.

```
Installing system [ Help ]

subiquity/2dev/apply_autoinstall_config:
subiquity/ad/apply_autoinstall_config:
subiquity/late/apply_autoinstall_config:
configuring apt
curtin command in-target
Installing System
executing curtin install initial step
executing curtin install partitioning step
curtin command install
configuring storage
  running 'curtin block-meta simple'
  curtin command block-meta
  removing previous storage devices
  configuring disk: disk-sda
  configuring partition: partition-0
  configuring partition: partition-1
  configuring format: format-0
  configuring partition: partition-2
  configuring lvm_volgroup: lvm_volgroup-0
  configuring lvm_partition: lvm_partition-0
  configuring format: format-1
  configuring mount: mount-1
  configuring mount: mount-0
executing curtin install extract step
curtin command install
  writing install sources to disk
  running 'curtin extract'
  curtin command extract
  acquiring and extracting image from cp:///tmp/tmpncu62m6y/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
  configuring installed system
  running 'curtin curthooks'
  curtin command curthooks
    configuring apt
    configuring apt
    installing missing packages
    configuring lsscsi service
    configuring raid (mdadm) service
    configuring NVMe over TCP
    installing kernel |
```

Step 9: Installation Complete

Once finished, the system prompted for a reboot.

```
Installation complete!

running 'curtin extract'
curtin command extract
acquiring and extracting image from cp:///tmp/tmpcu62m6y/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
configuring installed system
running 'curtin curthooks'
curtin command curthooks
configuring apt configuring apt
installing missing packages
configuring iscsi service
configuring raid (mdadm) service
configuring NVMe over TCP
installing kernel
setting up swap
apply networking config
writing etc/fstab
configuring multipath
updating packages on target system
configuring pollinate user-agent on target
configuring kernel crash dumps settings
final kernel configuration
configuring target system bootloader
installing grub to target devices
copying metadata from /cdrom
final system configuration
calculating extra packages to install
installing openssh-server
retrieving openssh-server
curtin command system-install
unpacking openssh-server
curtin command system-install
configuring cloud-init
downloading and installing security updates
curtin command in-target
restoring apt configuration
curtin command in-target
subiquity/late/run_builtin:
subiquity/late/run_user_supplied:
```

[View full log]
[Reboot Now]

6. Configuration Details

- Hostname: server01
- Filesystem: ext4
- Disk size: 40 GB
- Partition scheme: Guided LVM (boot partition + LVM volume group)
- SSH: Installed and enabled during setup
- Password authentication over SSH: Enabled

7. Verification Commands

Task 1 – Login

Logged into the server using the created user account.

Task 2 – Verify Hostname

Command: hostname

Result: returned server01

Task 3 – Verify IP Address

Command: ip addr

Result: confirmed IP address 10.0.2.15

Task 4 – Test Internet Connectivity

Command: ping -c 4 google.com

Result: 4 successful replies, 0% packet loss

```
deleon@server01:~$ hostname
server01
deleon@server01:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:a2:4e:84 brd ff:ff:ff:ff:ff:ff
    altname enx080027a24e84
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 85872sec preferred_lft 85872sec
    inet6 fd17:625c:f037:2:a00:27ff:fea2:4e84/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 85988sec preferred_lft 13988sec
    inet6 fe80::a00:27ff:fea2:4e84/64 scope link proto kernel_l1
        valid_lft forever preferred_lft forever
deleon@server01:~$ ping -c 4 google.com
PING google.com (172.217.26.78) 56(84) bytes of data.
64 bytes from kul08s14-in-f14.1e100.net (172.217.26.78): icmp_seq=1 ttl=64 time=7.60 ms
64 bytes from kul08s14-in-f14.1e100.net (172.217.26.78): icmp_seq=2 ttl=64 time=7.18 ms
64 bytes from kul08s14-in-f14.1e100.net (172.217.26.78): icmp_seq=3 ttl=64 time=7.43 ms
64 bytes from kul08s14-in-f14.1e100.net (172.217.26.78): icmp_seq=4 ttl=64 time=9.37 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3263ms
rtt min/avg/max/mdev = 7.182/7.895/9.370/0.864 ms
deleon@server01:~$ _
```

Task 5 – Update the Server

Command: `sudo apt update && sudo apt upgrade -y`

Result: packages updated successfully. One package (linux-firmware-amd-misc) failed to download from the regional mirror, resolved by switching to the global Ubuntu archive mirror.

```

File      Machine  View  Input  Devices  Help
-----
support  grub-pc-bin      linux-firmware-intel-graphics  linux-firmware-qualcomm-graphics  s3d-utils-udev
support-core-dump-handler  grub2-common      linux-firmware-intel-misc        linux-firmware-qualcomm-misc      software-properties-common
base-files      iproute2          linux-firmware-intel-wireless    linux-firmware-qualcomm-wireless  ubuntu-kernel-accessories
bind9-dnsssls   libapparmor1      linux-firmware-marvell-prestera  linux-firmware-realtek            ubuntu-minimal
bind9-dnsssls   libbind9-4        linux-firmware-marvell-wireless  linux-firmware-realtek-config     ubuntu-release-upgrader-core
bind9-lbns      libiscrypt2       linux-firmware-mediatek          linux-firmware-realtek            ubuntu-server
cloud-init      libpackagekit-glib2-1.0  linux-firmware-mellanox-spectrum  python3-apptools                  ubuntu-server-minimal
cloud-init-base libpki11s2-1.0-0   linux-firmware-misc              python3-distupgrade               ubuntu-standard
git2.1-packagekitgit1b-1.0  linux-firmware-amd-graphics  linux-firmware-pantheon           python3-software-properties
git2.1-packagekitgit1b-1.0  linux-firmware-amd-misc      linux-firmware-nvidia-graphics   python3-software-properties

Summary:
Upgrading: 53, Installing: 0, Removing: 0, Not Upgrading: 0
Download size: 717 MB
Space needed: 1.174 kB / 11.4 GB available

Det1: http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 motd-news-config all 14ubuntu6.1 [5,632 B]
Det2: http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 base-files amd64 14ubuntu1 [758 kB]
Det3: http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libcrypt2 amd64 1:12.0-ubuntu1 [571 kB]
Det4: http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 iproute2 amd64 6.19.0-ubuntu1 [1,198 kB]
Det5: http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libapparmor1 amd64 5.0.2-ubuntu176.04.1 [51.3 kB]
Det6: http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 ubuntu-minimal amd64 1:57.2-0 (9,534 B)
Det7: http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 apparmor amd64 5.0.2-ubuntu176.04.1 [380 kB]
Det8: http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 bind9-not-so amd64 1:9.20-24-ubuntu1 [49.8 kB]
Det9: http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 bind9-dnsssls amd64 1:9.20-24-ubuntu1 [184 kB]
Det10: http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 bind9-lbns amd64 1:9.20-24-ubuntu1 [1,308 kB]
Det11: http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 ubuntu-release-upgrader-core all 1:26.04.2 [24.9 kB]
Det12: http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 python3-distupgrade all 1:26.04.2 [102 kB]
Det13: http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 ubuntu-standard amd64 1:57.2-0 (9,534 B)
Det14: http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 python3-problem-report all 1:23.4-1-ubuntu1 [17.4 kB]
Det15: http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 python3-support all 1:23.4-1-ubuntu1 [56.4 kB]
Det16: http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 support-core-dump-handler all 1:23.4-1-ubuntu1 [9,674 B]
Det17: http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 support all 1:23.4-1-ubuntu1 [183.9 kB]
Det18: http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 cloud-init-base all 26.1-0ubuntu176.04.1 [587 kB]
Det19: http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 cloud-init all 26.1-0ubuntu176.04.1 [2,110 B]
Det20: http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libudm amd64 2.1.1-ubuntu1 [112.7 kB]
Det21: http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 librfid3 amd64 2.1.1-ubuntu1 [145 kB]
Det22: http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libpackagekit-glib2-1.0 amd64 1:3.4-3-ubuntu1 [1,21 kB]
Det23: http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 git2.1-packagekitgit1b-1.0 amd64 1:3.4-3-ubuntu1 [25.5 kB]
Det24: http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 grub-pc amd64 2.14-ubuntu1 [124 kB]
Det25: http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 grub-pc-bin amd64 2.14-ubuntu1 [1,325 kB]
Det26: http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 grub2-common amd64 2.14-ubuntu1 [2,586 kB]
Det27: http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libsg11s2-1.4B amd64 1:4B-ubuntu1 [120 kB]
Det28: http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-and-misc all 20260319.git217c6e4-0ubuntu1.1

408 Forbidden [IP: 202.79.180.254 80]
Det29: http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-and-graphics all 20260319.git217c6e4-0ubuntu1.1 [29.6 MB]
Det30: http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-and-graphics 14.3 MB/29.6 MB 48% 3,456 K/s 3min 175s
Det31: http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-and-graphics 16.6 MB/29.6 MB 55% 3,456 K/s 3min 165s
Det32: http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-broadcom-wireless all 20260319.git217c6e4-0ubuntu1.1 [11.2 MB]
Det33: http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-broadcom-wireless 2.263 MB/11.2 MB 20% 4,017 K/s 2min 45s

```

Task 6 – Verify SSH Service

Command: `systemctl status ssh`

Result: active (running)

```

deleon@server01:~$ systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Sat 2026-08-15 12:18:27 UTC; 21min ago
     Invocation: 4f0222c5920f4d29816423fd4cf1b8be
   TriggeredBy: ● ssh.socket
                Docs: man:sshd(8)
                   man:sshd_config(5)
   Main PID: 1296 (sshd)
     Tasks: 1 (limit: 1688)
    Memory: 1.6M (peak: 2.3M)
       CPU: 42ms
    CGroup: /system.slice/ssh.service
            └─1296 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Aug 15 12:18:27 server01 systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Aug 15 12:18:27 server01 sshd[1296]: Server listening on 0.0.0.0 port 22.
Aug 15 12:18:27 server01 sshd[1296]: Server listening on :: port 22.
Aug 15 12:18:27 server01 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
deleon@server01:~$

```

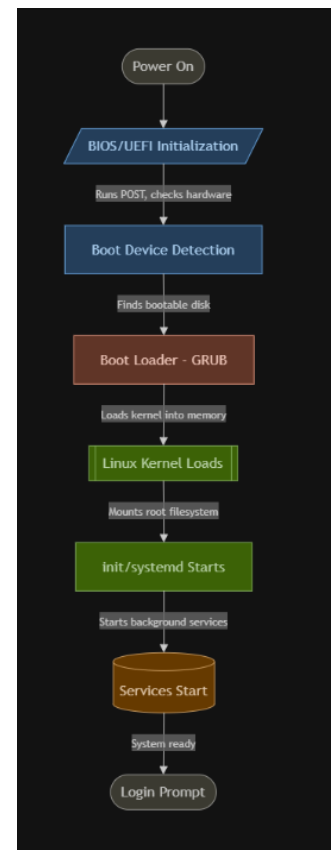
8. BIOS vs UEFI Comparison

BIOS and UEFI are the two types of firmware used to start up a computer before the operating system loads. BIOS is the older standard, limited to 2 TB disks, using the Master Boot Record for partitioning, and offering no built-in security checks during boot. UEFI is the modern replacement, supporting disks larger than 2 TB through the GUID Partition Table, booting faster, and including Secure Boot to block untrusted software from loading. UEFI also provides a graphical setup interface instead of BIOS's plain text menus. These improvements are the main reasons UEFI has replaced BIOS on nearly all computers built in the last several years, including being a requirement for Windows 11.

The full comparison table and detailed explanation are available in [\[BIOS_vs_UEFI.pdf \]](#)

9. Boot Process Flowchart

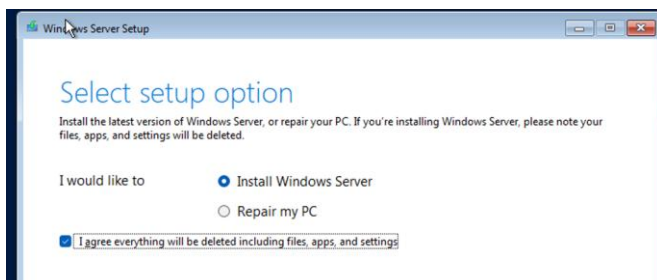
The Ubuntu boot process follows eight stages: Power On, BIOS/UEFI Initialization, Boot Device Detection, Boot Loader (GRUB), Linux Kernel Loads, init/systemd Starts, Services Start, and Login Prompt. [\[diagrams\BootProcessFlowchart.png \]](#)



10. Windows Server Installation Summary

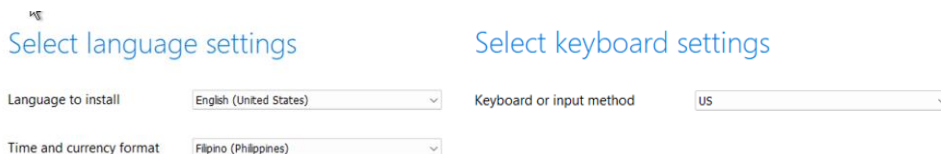
Step 1: Setup Option

Selected "Install Windows Server" to begin a fresh installation.



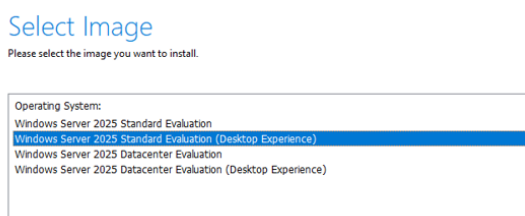
Step 2: Language, Time, and Keyboard

Set to English (United States), US keyboard.



Step 3: Edition Selection

Selected "Windows Server 2025 Standard Evaluation (Desktop Experience)" for the full graphical environment.



Step 4: License Terms

Accepted the Microsoft license agreement.

Applicable notices and license terms

Please read this so you know what you're agreeing to.

released version of this software will be subject to all the terms of this agreement at (aka.ms/useterms).

Notwithstanding anything to the contrary in this agreement, **previews are provided "AS IS," and no warranty, implied or express (including the Limited Warranty), applies to these versions. By installing previews on your device, you may void or impact your device warranty and may not be entitled to support from the manufacturer of your device or network operator, if applicable.** Microsoft is not responsible for any damage thereby caused to you. Microsoft may not provide support services for previews. If you provide Microsoft comments, suggestions or other feedback about the preview ("submission"), you grant Microsoft and its partners rights to use the submission in any way and for any purpose.

EULAID:Sept2020_DCSTD_EN-US

Microsoft Support Legal

Decline Accept

Step 5: Disk Selection

Selected the unallocated 50 GB virtual disk for a clean custom installation.

Select location to install Windows Server

Refresh

Load Driver

Bring Disk Online

Delete Partition

Format Partition

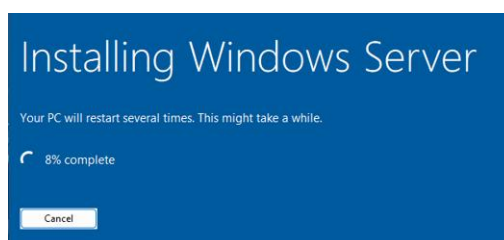
Create Partition

Extend Partition

Name	Total Size	Free Space	Type
Disk 0 Unallocated Space	50.0 GB	50.0 GB	Unallocated Space

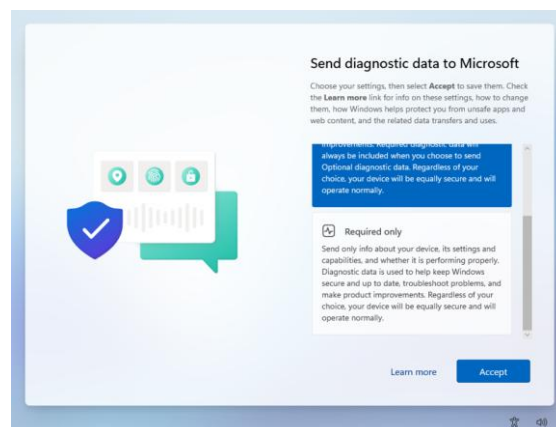
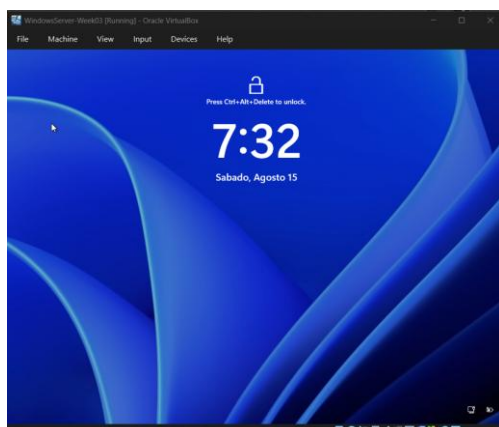
Step 6: Installation Progress

Windows Setup partitioned the disk and installed the system, rebooting automatically to complete setup.



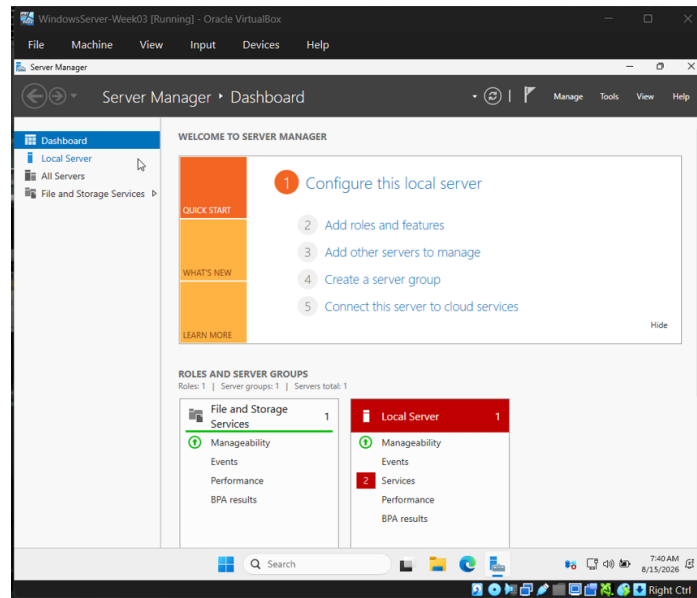
Step 7: First Login

Logged in for the first time using Ctrl+Alt+Del and the configured Administrator password.



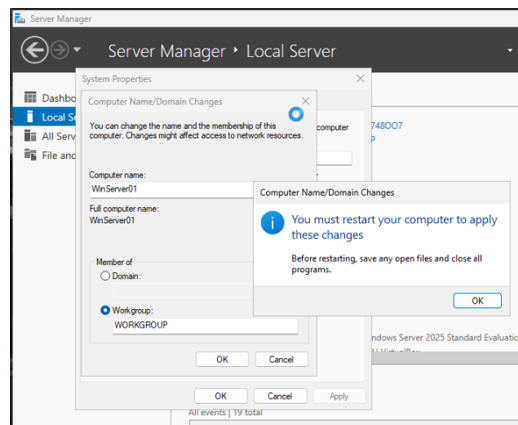
Step 8: Server Manager Dashboard

Confirmed successful installation via the Server Manager dashboard that opens automatically on login.



Step 9: Computer Name Assignment

Renamed the computer from its default auto-generated name to WinServer01 through System Properties.



11. OS Comparison Table

Aspect	Windows Server	Ubuntu Server	Rocky Linux
Developer	Microsoft	Canonical	Rocky Enterprise Software Foundation
Licensing	Paid, requires a purchased license	Free and open source	Free and open source
Release Model	New version every few years	New version every 6 months, LTS every 2 years	Follows Red Hat Enterprise Linux release cycle
User Interface	Graphical interface by default	Command line by default, GUI optional	Command line by default, GUI optional
Package Management	PowerShell, Server Manager, MSI installers	APT (apt install, apt update)	DNF/YUM (dnf install, dnf update)

Security	Strong security, bigger malware target due to popularity	Frequent updates, smaller attack surface	Very security focused, strict compliance use
Performance	Uses more resources due to GUI	Lightweight	Lightweight
Stability	Stable, sometimes needs reboots after updates	Very stable, rarely needs reboot for updates	Extremely stable, built for long term use
Typical Enterprise Use Cases	Active Directory, .NET apps, SQL Server	Web hosting, cloud servers, Docker	Data centers, banks, RHEL replacements
Cost Over Time	Higher due to licensing and CALs	Free	Free
Advantages	Familiar interface, strong Microsoft ecosystem	Free, lightweight, huge community	Free, extremely stable, RHEL compatible
Disadvantages	Expensive licensing, uses more resources	Command line learning curve	Smaller community

Discussion

Choosing between Windows Server, Ubuntu Server, and Rocky Linux really comes down to what a company already uses and what they are trying to build. Each one has a clear strength that makes it the better choice in different situations.

Windows Server is the natural pick for companies that already run mostly on Microsoft products. If a business uses Active Directory to manage user accounts, or runs applications built with .NET, or depends on SQL Server for their databases, Windows Server fits right in without much extra setup. Its biggest advantage is familiarity. The downside is cost, since Windows Server needs a paid license, and many setups also need Client Access Licenses for each user or device connecting to the server.

Ubuntu Server takes the opposite approach. It is completely free, and its biggest strength is how often it gets updated and how active its community is. It fits naturally into modern web hosting, cloud computing, and container-based systems like Docker. It uses fewer system resources compared to Windows Server since it does not run a full graphical interface by default. The tradeoff is that it takes more effort to learn, especially for someone who has never used a command line before.

Rocky Linux exists mainly for companies that need long term stability more than anything else. It was created to fill the gap left when CentOS changed its update model, and it follows the same release pattern as Red Hat Enterprise Linux, which is widely trusted in large companies, banks, and government systems. This makes it a strong choice for industries where downtime is costly or where compliance rules require a stable, well tested operating system.

In the end, there is no single best choice among the three, it depends on the situation. A company deep in the Microsoft ecosystem will likely stick with Windows Server despite the cost. A startup building a modern cloud-based product will probably lean toward Ubuntu Server because of its

flexibility and strong community support. And a business that needs rock solid stability for years without much change will often choose Rocky Linux.

12. Problems Encountered

- The Ubuntu Server virtual machine crashed unexpectedly right after the user profile setup screen during the first installation attempt, requiring a full restart of the installation process.
- During the system update step (`sudo apt update && sudo apt upgrade -y`), one package (`linux-firmware-amd-misc`) failed to download from the regional Ubuntu mirror (`ph.archive.ubuntu.com`), returning a 403 Forbidden error.

13. Solutions Implemented

- Restarted the virtual machine and repeated the Ubuntu Server installation process from the beginning. On the second attempt, care was taken to navigate installer screens using Tab rather than arrow keys to avoid input issues.
- Resolved the package download failure by switching the APT source from the regional Philippine mirror to the global Ubuntu archive mirror using `sed`, then re-running `apt update` and `apt upgrade`. The failed package was confirmed to be non-essential AMD GPU firmware, not required in a virtualized environment.

14. Lessons Learned

This activity provided hands-on experience with enterprise server deployment from start to finish. Working through Ubuntu Server's text-based installer built familiarity with Linux server administration, including disk partitioning with LVM, SSH configuration, and command-line verification tasks. Troubleshooting the VM crash and the mirror download failure reinforced the importance of patience and systematic problem-solving in system administration work. Comparing BIOS and UEFI, along with Windows Server, Ubuntu Server, and Rocky Linux, also built a clearer understanding of how infrastructure decisions depend on an organization's existing ecosystem, budget, and stability requirements.

15. References

- Ubuntu Server official documentation: <https://ubuntu.com/server/docs>
- Canonical, Ubuntu 26.04 LTS Release Notes
- Microsoft Windows Server 2025 Evaluation Center